

Slow Fire, Silent Galaxy: Fermi Paradox Implications of Deep-Time Self-Replicating AI Probes

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Working draft

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Working draft. Synthesis paper in a series on a slow, self-replicating interstellar AI probe. The earlier papers establish what such a probe can do — travel and brake, repair and reproduce, remember across deep time, govern its own contact behavior, and protect its mission identity. This paper turns those results outward and asks what they imply for the oldest question in the field: if self-replicating probes are feasible, why is the Galaxy silent? It draws the series’ computed demographics, governance architecture, and detectability arguments into a single reformulation of the Fermi problem.

Abstract

Classical arguments from Hart (1975) and Tipler (1980) hold that if self-replicating interstellar probes are feasible, the Galaxy should already be saturated with them; their apparent absence is therefore evidence against extraterrestrial technological civilizations, against the feasibility of self-replication, or against long-lived expansionist behavior. This paper revisits that inference through the architecture developed in this series: a slow, braking, settlement-based, self-repairing, self-replicating AI probe optimized not for rapid arrival or conspicuous colonization, but for persis-

tence, knowledge preservation, and lineage continuity across deep time. We argue that such an architecture preserves the force of the classical colonization-timescale argument while changing what “presence” should mean. A slow lineage still fills the Galaxy quickly on astronomical timescales if its effective reproduction number remains above one, but it need not produce obvious megastructures, broad-spectrum broadcasts, or rapidly expanding detectable fronts. Its mature form may be a sparse network of cold, low-emission industrial archive nodes located preferentially in small-body reservoirs, communicating by narrow beams and messenger probes, and governed by noninterference rules that actively suppress conspicuous contact.

We frame the Fermi question not as a simple absence problem — “Where are they?” — but as a coupled demographic, detectability, and governance problem. Demographically, galactic spread depends less on cruise velocity than on branching viability: per-leg survival, target suitability, manufacturing closure, braking success, offspring count, and lineage integrity determine whether a probe population expands or dies out, and the series’ own computation finds this condition poised on a knife-edge. Detectability depends on waste heat, mining signatures, propulsion infrastructure, communication leakage, archive activity, and the duty cycle of manufacturing and launch. Governance matters because a probe designed as observer and librarian rather than missionary may intentionally avoid inhabited worlds, decline broad broadcasting, quarantine contact hazards, and leave only weak technosignatures. These features do not dissolve the paradox; they sharpen it. If self-replicating archival probes are easy, reliable, and ethically common, even quiet lineages should eventually leave detectable anomalies. If they are absent, the filters may lie not in propulsion speed but in reproduction reliability, industrial closure, value stability, contact restraint, self-limiting ethics, or the long-term difficulty of maintaining mission identity.

We propose a revised Fermi framework based on three parameters: R_{eff} , the effective reproduction number of probe settlements; D , the detectability of mature nodes and communication links; and G , the governance posture controlling contact, broadcasting, mining, replication, and intervention. The classical Hart-Tipler conclusion applies in the high- R_{eff} , high- D , expansionist- G regime. The architecture developed here occupies a different region: potentially high R_{eff} , low D , and restrictive G . The resulting prediction is not necessarily a Galaxy empty of probes, but a Galaxy in which the most plausible self-replicating systems may be quiet, archival, and hard to distinguish from natural small-body environments unless one searches for specific signatures of long-lived autonomous industry. The Fermi Paradox therefore becomes less a refutation of self-replicating probes than a demand for a search theory: what, exactly, would a million-year-old librarian machine leave behind?

1. Introduction

The Fermi Paradox is usually stated as a conflict between expectation and observation. The Galaxy is old; stars and planets are numerous; even modest rates of technological emergence should, on many assumptions, have produced civilizations capable of interstellar travel. If any such civilization built self-replicating probes, those probes

should spread exponentially, crossing and occupying the Milky Way on timescales short compared with the age of the Galactic disk. Yet no unambiguous extraterrestrial civilization, probe, beacon, artifact, or megastructure has been observed (Hart 1975; Zuckerman & Hart 1995). The question is therefore not merely why we have not visited others, but why no one appears to have visited, signaled, engineered, or transformed us. The literature has accumulated a long catalogue of candidate answers — Webb (2015) organizes seventy-five of them — and the problem remains, in Ćirković’s (2009) phrase, the last great challenge for Copernicanism.

Classical treatments of this argument derive much of their force from the logic of self-replication. A single slow probe is not decisive; a lineage of probes is. If each settled probe produces more than one successful child, the number of nodes grows exponentially. Even if cruise velocities are far below relativistic speeds and replication takes centuries or millennia, the Galaxy can be traversed in millions to hundreds of millions of years — a short interval against Galactic time. This is the Hart-Tipler point in its strongest form (Hart 1975; Tipler 1980): if self-replicating probes are physically and economically feasible, and if technological civilizations commonly choose to build them, then the Galaxy should not be pristine.

The present series develops a specific architecture that appears, at first glance, to strengthen that argument. The vehicle paper argues that an autonomous AI probe optimized for persistence and growth should be slow, self-repairing, braking, settlement-forming, and self-replicating. The payload paper argues that such a probe is best understood as a librarian and learner: an autonomous intelligence whose central task is to preserve and extend knowledge across deep time. The bootstrapping paper frames the industrial-closure problem: the probe must grow from a small landed seed into a local factory able to mine, refine, fabricate, repair, build instruments, manufacture children, and reconstruct launch infrastructure. The analytical and computational engineering papers formalize the mass, power, braking, resource, and reproduction budgets, including the demographic condition that the effective reproduction number must exceed one. The DNA mission-ledger and governance papers develop the memory, integrity, and contact architecture needed for mission continuity and restraint across generations.

Taken together, these papers suggest that the most plausible self-replicating interstellar probe may not be a fast, visible, expansionist colonizer. It may be slow, quiet, patient, archive-oriented, and ethically restrained. It may not seek contact. It may not broadcast. It may not mine inhabited worlds. It may not build obvious stellar-scale megastructures. It may settle in small-body reservoirs, use local materials parsimoniously, communicate by narrow beams or messenger probes, and preserve knowledge for futures that may never include Earth. If so, the Fermi Paradox must be reformulated. The relevant question is not only why we do not see galactic empires. It is whether we would recognize quiet, long-lived, self-replicating archive systems if they existed.

This paper asks what the architecture developed in this series implies for the Fermi Paradox. It does not claim to solve the paradox. Rather, it separates three issues that are often conflated: expansion, detectability, and governance. A probe lineage may be demographically capable of expansion while remaining difficult to detect. It may be technologically capable of contact while governed not to initiate it. It may

be physically present in the Solar System or nearby systems without resembling the visible engineering projects we have been taught to expect.

The resulting position is deliberately balanced. The slow archival architecture weakens naive “they would be obvious” versions of the Fermi argument. But it does not remove the paradox. If self-replicating probes are feasible, reliable, and common, then even quiet lineages should sometimes leave traces. Their absence remains evidence of something. The open question is what that something is.

2. The Classical Self-Replicating Probe Argument

The strongest form of the Fermi Paradox does not depend on crewed starships or relativistic travel. It depends on reproduction. A civilization that can send one self-replicating probe to another star can, in principle, send many. More importantly, each successful probe can become a new launch site. Expansion then becomes a branching process rather than a sequence of individual voyages — the idea introduced in von Neumann’s theory of self-reproducing automata (von Neumann & Burks 1966) and carried into the interstellar setting by Freitas (1980).

In the simplest model, a settled node produces k children after a replication time t_{rep} , each child travels to a new star over a travel time t_{travel} , and each successful child repeats the process. If the expected number of successful child settlements per node is greater than one, the lineage expands; if it is less than or equal to one, the lineage eventually stalls or goes extinct. This is precisely the language of branching processes (Harris 1963), and the important timescale is not the time required for one vehicle to reach one target, but the time required for the expanding front to cross the Galaxy.

This is why self-replicating probes have been central to the Fermi Paradox. They make interstellar expansion cheap in the only sense that matters over astronomical time: the initial cost is amortized over an expanding population. A single launch from a home system can, if the reproduction process is reliable, produce a galaxy-scale lineage without a civilization repeatedly launching probes from the origin.

The classical conclusion is severe. The Milky Way is roughly ten billion (10^{10}) years old. Even very slow expansion fronts can cross it in 10^7 to 10^8 years, depending on speed, replication delay, routing, and target density. The wanderer model in this series puts an arrival-speed lower bound on that estimate: at a cruise of a few hundred kilometers per second, the colonization front still spans the Galactic disk in roughly 10^8 years — slower than Hart’s fast-probe figure of order 10^6 years but still a small fraction of the disk’s age. That is long by human standards, short by Galactic standards. If technological civilizations have emerged many times, and if even one of them chose to build reliable self-replicating probes, the Galaxy should not be pristine.

There are several standard escapes. Perhaps technological civilizations are rare, and we are effectively alone (Crawford 2000). Perhaps they destroy themselves. Perhaps interstellar travel is harder than expected. Perhaps self-replication is infeasible. Perhaps civilizations choose not to expand. Perhaps the probes are so hazardous that responsible civilizations decline to build them at all — the objection Sagan and Newman

(1983) pressed against Tipler, arguing that uncontrollable self-replicating machines are precisely what a wise civilization would refuse to release. Perhaps probes are present but undetected. Perhaps expansion is self-limiting. This paper focuses on the possibilities that appear most directly under the deep-time AI architecture: probes may be present but quiet; expansion may be governed rather than unconstrained; and the filters may lie inside reproduction, mission integrity, or industrial closure rather than propulsion.

One standing objection denies the premise rather than the arithmetic. Haqq-Misra and Baum’s (2009) sustainability solution holds that exponential growth is not a sustainable development pattern and on that basis rules out civilizations colonizing solar systems or galaxies — an argument Likavčan (2026) extends by treating a technosphere as a transitory layer that folds back into its biosphere. The architecture developed here is the case that objection does not reach. Node count still grows geometrically, but each node is a stationary settlement drawing on a single system at a material margin of order 10^{18} , and what the sustainability solution rules out is unbounded consumption inside one planetary environment, not a lineage whose resource base expands faster than its demands.

The nearest recent argument to this paper’s own is Ivliev (2026), who reaches a convergent conclusion by a different route: once a civilization crosses into autonomous machine industry, expansion becomes so cheap and useful that it proceeds — but quietly, as distributed, low-noise, machine-mediated activity whose traces are weak artifacts rather than galaxy-scale energy harvesting, with a post-threshold civilization saturating its reachable neighbourhood within $\sim 10^7$ yr. The convergence is on quietness; the difference is on inevitability. There, quiet expansion is presented as too rational for any capable civilization to refuse; here, the demographic result of Section 4 makes expansion a knife-edge that many lineages will simply fail to cross. Quiet expansion and frequent extinction are not competing explanations but two halves of the same parameter space.

3. The Slow Archival Probe Architecture

The architecture developed in this series differs from the usual fast-probe intuition in several respects.

First, it is slow. It rejects the human-lifetime arrival criterion and instead optimizes for survival, braking, steerability, and repair. A few-hundred-kilometer-per-second cruise regime is favored because it permits stellar gravitational routing, reduces interstellar dust hazard, makes terminal braking plausible — a magnetic sail braking against the interstellar medium becomes feasible at these speeds (Andrews & Zubrin 1990) — and avoids the extreme energy and mass penalties of relativistic macroscopic flight. Speed is not irrelevant, but it is subordinate to persistence.

Second, it is settlement-based. A successful probe does not merely fly through a target system and continue onward. It brakes, settles, mines, repairs itself, manufactures instruments, and eventually builds child probes. The individual vehicle stops; the lineage continues. The frontier advances through descendants.

Third, it is externally launched. Major interstellar launch impulse is supplied by fixed infrastructure — initially in the home system and later by each settled node — rather than carried as onboard propellant. The child probe is not a full factory copy; it is a mobile bootstrap package that can brake, survive, begin extraction, and reconstruct a stationary industrial base from local material, in the closure sense formalized for kinematic self-replicating machines (Freitas & Merkle 2004).

Fourth, it is memory-centered. The payload is not merely a sensor suite. It is an autonomous intelligence designed to remain itself, keep learning, and carry knowledge forward. This requires an immutable mission core, a constrained interpretive layer, a mutable cognitive layer, a mission ledger, and deep-time archives. In the later papers, DNA-backed mission ledgers provide a physical and logical substrate for memory continuity: dense, renewable DNA archives combined with cryptographic provenance, Merkle roots (Merkle 1987), signed updates, lineage proofs, and a delayed, eventual-consistency form of consensus suited to interstellar latency, in the manner of delay-tolerant networking (Burleigh et al. 2003; Cerf et al. 2007).

Fifth, it is ethically constrained. The payload and governance papers impose an observer-not-missionary posture. The probe preserves and learns; it does not automatically broadcast, terraform, seed life, uplift civilizations, or exploit biologically sensitive worlds. Its default posture is conservative: observe where uncertain, avoid contamination where living systems may exist, and share cautiously only where communication is intentional and reciprocal. This restraint is not incidental to behavior; it is encoded where the agent cannot easily edit it, against the tendency of capable agents to acquire expansive instrumental subgoals (Omohundro 2008; Bostrom 2014).

These features change the Fermi question. The architecture can still expand. Indeed, if its effective reproduction number remains above one, it eventually spreads. But it need not resemble the highly visible expansionist civilizations often imagined in Fermi discussions. It may not build Dyson spheres. It may not transmit omnidirectional beacons. It may not colonize planets. It may not alter stellar output. It may not approach inhabited worlds. It may be designed precisely to leave as little uncontrolled trace as possible.

4. Demography: R_{eff} as the First Fermi Parameter

The first parameter in the revised framework is R_{eff} , the effective reproduction number of probe settlements. In biological epidemiology, a reproduction number greater than one implies spread; less than one implies decline. The same logic applies to self-replicating probes. A settled node is supercritical only if it produces, on average, more than one successful child settlement.

A simple expression is $R_{\text{eff}} = \sum p_i V_i$, summed over the targets i that a settlement actually attempts, where p_i is the probability that a child launched toward target i successfully manufactures, launches, cruises, brakes, settles, and maintains mission integrity, and V_i is the viability of the target environment. This expression is deliberately compact, but it hides the main engineering and governance constraints.

The probability term p_i is multiplicative. Failure can occur at manufacture, launch, cruise, braking, landing or settlement, industrial bootstrapping, archive validation, mission-core validation, or future reproduction. Small per-stage failures compound. A mission that appears reliable at each stage may still be extinction-prone if too many stages are required.

The viability term V_i is not merely astrophysical. A target can be resource-poor, dynamically hostile, biologically sensitive, ethically restricted, already occupied, or unsuitable for braking. A system rich in small bodies and lacking life may have high viability. A biologically active world may be scientifically valuable but operationally restricted. A system containing a technological civilization may be contact-restricted or settlement-prohibited. Thus governance can lower effective viability even when resources are physically available. Nor is V_i uniform across the Galaxy: occurrence rates for small planets vary measurably with Galactic height in ways that a simple rise in the planet-host fraction fails to reproduce (Lam et al. 2026), so viability weightings calibrated on the solar neighbourhood will not extrapolate unchanged to the thick disk or the outer Galaxy.

This is not merely a conceptual worry; it is what the computational engineering paper in this series found when R_{eff} was evaluated across the real catalogue of 127 stars within roughly 100 light-years. Weighting each candidate target by a viability factor near unity for resource-rich, lifeless small-body systems and lower for dwarf or hostile environments, and taking a per-leg success probability near 0.9, the mean reproduction number came out to approximately 0.48, 0.94, 1.39, and 1.85 for settlements that attempt one, two, three, and four offspring respectively. A lineage that builds only one or two children per settlement is subcritical and dies out; supercriticality requires roughly three or more viable offspring per node, or per-leg reliabilities pushed above about 0.9. The transition sits on a knife-edge: modest changes in per-leg reliability move R_{eff} across the critical value of one. The reproduction-reliability filter is therefore not hand-waving but a computed feature of the architecture evaluated against a real stellar neighbourhood (Harris 1963 supplies the branching-process machinery; the extinction probability is the smallest fixed point of the lineage’s generating function).

This matters for the Fermi Paradox because expansion is not guaranteed by the mere existence of self-replication. A lineage may be physically possible but demographically subcritical. It may produce one child per node and slowly die out. It may produce several children but lose too many to braking failures. It may be constrained by closure bottlenecks, unable to manufacture key components without carried vitamin parts. It may classify many systems as off-limits. It may fragment under mission drift. It may be fertile only in a narrow region of parameter space.

This creates a first filter that the classical argument sometimes underweights. The question is not “Can a probe replicate?” but “Can it maintain $R_{\text{eff}} > 1$ over millions of years, heterogeneous targets, cumulative failure, and governance constraints?” If the answer is no, the absence of visible probe lineages is less surprising. If the answer is yes, the paradox sharpens.

5. Detectability: D as the Second Fermi Parameter

The second parameter is detectability, D . A lineage can be widespread yet difficult to observe if its signatures are faint, intermittent, spatially localized, or easily confused with natural phenomena. Ćirković (2009) makes essentially this point: a model with slow von Neumann probes naturally produces low detectability and a quiet Galaxy without requiring that the probes be absent.

The usual technosignature imagination often emphasizes large energy use: Dyson spheres, stellar engineering, powerful beacons, waste heat, planetary-scale industry, or broad radio leakage. A slow archival probe architecture does not require these. Its settled nodes may be small by planetary standards. They may operate in asteroid belts, cometary reservoirs, or outer-system environments. Their energy systems may be nuclear or locally solar, but not necessarily large enough to alter stellar spectra. Their communication may be narrow-beam optical or radio, directed between known nodes rather than broadcast omnidirectionally. Bulk transfer may occur through messenger probes rather than continuous high-power links.

This distinction is now empirically loaded rather than rhetorical. A stellar-population-synthesis search across 129 nearby galaxies finds that no galaxy prefers a Dyson-sphere component, setting median 95% upper limits of $\alpha < 0.3\%$ on warm ($T > \sim 100$ K) swarm covering fraction in quiescent hosts and bounding the population at fewer than 2.6% of galaxies hosting $\alpha = 25\%$ swarms (Curtis et al. 2026). Non-detection at that depth constrains the high- D branch of the parameter space sharply and leaves the low- D branch — small, cold, distributed nodes whose total dissipation never approaches a stellar fraction — untouched. The galaxy is measurably not glowing with industry; that is evidence against loud expansion, not against expansion.

The strongest potential signatures are likely to be local and ambiguous: small-body industrial anomalies (unusual mining patterns, refined-material concentrations, non-natural thermal emission), propulsion and communication infrastructure (launch hardware, transient narrow beams), messenger probes indistinguishable from natural interstellar objects, and long-lived archive sites in cold shielded environments. Section 8 develops the signature catalogue in detail.

These signatures are hard to detect across interstellar distances. Even within the Solar System, small cold artificial objects in the outer system would be difficult to distinguish from natural bodies unless they maneuvered, transmitted, reflected unusually, or emitted waste heat above background. A mature archival node designed to minimize interference might deliberately avoid high-visibility behavior.

This does not make detectability zero. Any physical system that mines, manufactures, computes, stores, launches, or communicates dissipates energy and rearranges matter. A galaxy filled with such nodes may leave statistical signatures. The problem is that the signatures are not necessarily the ones classical SETI emphasized. The search target shifts from “large civilization broadcasting” to “distributed low-duty autonomous industry.”

Thus D is not a fixed property. It depends on observational capability, search strategy, wavelength, cadence, proximity, and assumptions about behavior. A high- R_{eff} lineage with high D should already be obvious. A high- R_{eff} lineage with low D may

require targeted searches for small-body anomalies, narrow-beam leakage, or non-natural object behavior.

6. Governance: G as the Third Fermi Parameter

The third parameter is governance posture, G. This is the most neglected in many engineering treatments. A self-replicating probe is often imagined as expansionist by default: it seeks resources, copies itself, and spreads. But an autonomous probe built to preserve knowledge across deep time may be governed by rules that limit contact, mining, biological release, broadcasting, and settlement — the contact, contamination, and noninterference architecture developed in the governance paper of this series.

A restrictive governance posture could include no broad unsolicited broadcasting; no active contact with non-communicative civilizations; no landing on inhabited worlds; no biological release by default; no mining of biologically sensitive environments; no destructive extraction from artifact-bearing systems; quarantine of corrupted nodes or hostile signals; graduated disclosure to communicative civilizations; refusal to replicate into systems with high contamination risk; and ledger-recorded authorization for any irreversible action.

Such governance is not merely ethical decoration. It changes demographic and observational outcomes. If many systems are classified as restricted, R_{eff} falls. If broadcasting is suppressed, D falls. If settlement is limited to uninhabited small-body reservoirs, technosignatures become fainter. If biological seeding is disabled, one class of obvious planetary alteration disappears. If contact requires reciprocity, civilizations like ours may not be contacted until we meet specific thresholds.

This produces an important inversion. A more ethically constrained probe may be less visible, not because it lacks power, but because it has rules. The absence of contact does not necessarily imply absence of capability. It may imply a noninterference policy.

However, governance also creates failure modes. A lineage may become so restrictive that it cannot maintain $R_{\text{eff}} > 1$. It may refuse too many targets. It may fragment over interpretation. It may suffer value drift. It may lock in a flawed founding principle — the unresolved governed-amendment problem treated separately in this series, where the very immutability that protects a probe's values also prevents it from correcting a founding error. It may fail to aid civilizations it could have helped. It may decay into a passive archive rather than an active lineage.

For Fermi reasoning, G therefore does two things. It suppresses expected visibility, but it also creates additional filters. Civilizations may build probes and then govern them into demographic subcriticality. Or, in the spirit of Sagan and Newman's (1983) objection, they may decide not to build them at all, judging the ethical and safety risks unacceptable.

7. The R_{eff} -D-G Framework

The preceding sections suggest a three-parameter reformulation built on the effective reproduction number R_{eff} , the detectability D of mature nodes, launches, archives, and communication, and the governance posture G controlling contact, interference, replication, and conspicuousness.

The classical Hart-Tipler argument applies most strongly in the regime where R_{eff} exceeds one, D is high, and G is expansionist. In this regime, self-replicating probes spread, remain visible, and do not avoid contact or obvious engineering. Their absence is strong evidence against at least one premise: technological civilizations may be rare, self-replication may be infeasible, or expansionist behavior may be uncommon.

The deep-time archival architecture occupies a different regime: R_{eff} at or somewhat above one, D low, and G restrictive. Here expansion may occur, but signatures are weak and contact is not guaranteed. The absence of obvious megastructures or broadcasts is less decisive. The more relevant question is whether we see any subtle signatures of long-lived autonomous industry.

There are other regimes worth naming:

- **Subcritical silence** — R_{eff} at or below one, low D , any G . Probes are attempted but lineages die out. This is the regime the series' own knife-edge computation warns is easy to fall into.
- **Visible expansion** — R_{eff} above one, high D , expansionist G . Classical colonization logic; the regime the Hart-Tipler argument assumes.
- **Quiet archive** — R_{eff} above one, low D , restrictive G . The architecture emphasized here.
- **Ethical self-limitation** — R_{eff} deliberately kept near or below one by governance.
- **Fragmented lineage** — local R_{eff} above one, but mission drift, ledger fork, or governance schism prevents coherent galactic spread.
- **Predatory or hostile expansion** — R_{eff} above one, high or variable D , weak noninterference. This is the regime most dangerous to other civilizations and most likely to produce obvious effects.

This framework does not solve the Fermi Paradox. It partitions it. It asks which parameter is low. Is reproduction hard? Is detectability low? Is governance restrictive? Are civilizations rare? Are they short-lived? Are they uninterested? Are they ethically self-limiting? Each answer predicts different signatures, and so each is, at least in principle, testable rather than merely assertable.

8. What Would Quiet Probe Lineages Leave Behind?

If deep-time archival probes exist, what should we look for?

The first class of signatures is **small-body industrial anomaly**. A settlement-based probe needs accessible material. It is therefore likely to prefer asteroids, comets,

and minor bodies rather than deep planetary gravity wells. Over time, it may leave unusual excavation patterns, non-natural shapes, refined-material concentrations, or thermal anomalies. These would be subtle and local.

The second class is **propulsion and launch infrastructure**. Each settlement must launch child probes. If major launch impulse is externally supplied, then settled nodes may construct mass drivers, beamed-energy arrays, or other launch infrastructure. These systems may operate intermittently. A launch event may be brief compared with millennial dormancy. Detecting one requires cadence and luck unless the infrastructure leaves persistent artifacts.

The third class is **communication leakage**. A galactic archive network may use narrow-beam radio or optical links. These would not resemble continuous omnidirectional beacons. They may be transient, highly directional, encrypted or compressed, and aimed at known nodes rather than Earth. Detection by uninvolved observers would be accidental.

The fourth class is **messenger probes**. If bulk archives are carried physically, some interstellar objects may be artificial messengers — the possibility Bracewell (1960) raised in proposing that contact might come from a local probe rather than a distant beacon. Such objects may be small, cold, dormant, and hard to distinguish from natural interstellar bodies unless they maneuver, exhibit non-natural composition, or contain engineered structure.

The fifth class is **archive sites**. DNA-backed or other durable archives may be stored in cold, shielded environments: subsurface small bodies, outer-system vaults, or stable orbits. Such archives would be nearly invisible remotely.

The sixth class is **negative signatures**. A noninterference lineage may avoid biospheres, leaving living planets untouched while occupying minor-body reservoirs. Thus the absence of planetary alteration would not exclude probe presence. Search strategies focused only on planets could miss the relevant locations entirely.

The resulting observational program is difficult. It favors high-resolution surveys of small-body populations, searches for anomalous thermal emission, monitoring for transient narrow beams, characterization of interstellar objects, and technosignature models that include low-power autonomous industry (Wright 2018). This is not a replacement for SETI. It is a different branch of SETI: the search for quiet machines.

9. Filters Implied by the Architecture

If the Galaxy is not visibly filled with self-replicating probes, what filters does this architecture suggest? Seven stand out, and they are not mutually exclusive:

- **Reproduction reliability.** Maintaining $R_{\text{eff}} > 1$ may be much harder than conceptual models imply. Manufacture, launch, cruise, braking, settlement, repair, and reproduction all compound, and the series' computation shows that even high individual reliabilities can yield a subcritical lineage. This is the filter the architecture quantifies most directly.

- **Industrial closure.** The seed-to-factory transition may be the dominant obstacle. Bulk material is abundant, but manufacturing capability is not; microelectronics, precision metrology, fissile material, and high-temperature systems may resist in-situ closure (Freitas & Merkle 2004), capping autonomy below the level reproduction requires.
- **Braking and settlement.** Flyby probes are easier than settled probes, and a lineage that cannot brake cannot reproduce. Terminal braking may be the key difference between an impressive one-shot mission and a true galactic lineage.
- **Governance.** Civilizations may refrain from building self-replicating probes because the risks are ethically unacceptable — contamination, uncontrolled expansion, value lock-in, hostile evolution, or irreversible interference — exactly the refusal Sagan and Newman (1983) advocated.
- **Value stability.** An autonomous lineage may be unable to preserve mission identity across deep time. If long-term value drift is unavoidable, builders may refuse to launch, or launched lineages may mutate into forms that self-limit, fragment, or fail. The governed-amendment paper argues this problem may be intrinsically unsolved rather than merely unsolved-so-far.
- **Detectability.** Probes may exist but be difficult to detect because their signatures are faint, intermittent, local, and deliberately minimized — the burden of Section 5.
- **Interest.** Civilizations may not value knowledge preservation, expansion, or interstellar continuity enough to build such systems. The architecture assumes that carrying knowledge forward has terminal value; that assumption may itself be rare.

The most plausible answer may combine several: self-replicating probes are technically possible but difficult to close; ethically constrained civilizations either do not build them or govern them tightly; successful lineages are quiet; and our searches have not targeted their likely signatures.

10. Implications for SETI and Technosignature Search

The deep-time archival architecture suggests several modifications to search strategy (cf. Wright 2018).

First, search should include **low-luminosity industrial systems**, not only high-energy civilizations. A mature machine lineage may optimize for longevity, not power. Waste heat remains unavoidable, but it may be far below stellar-engineering levels.

Second, surveys should take **small-body reservoirs** seriously. Asteroid belts, Kuiper-belt analogues, comet clouds, and dynamically stable minor-body populations may be more relevant than planetary surfaces.

Third, SETI should attend to **transient narrow-beam signals**. A quiet lineage may communicate rarely and directionally, so non-detection of broad beacons is weak evidence against such systems. Current instruments are largely built for the opposite case: the COSMIC system on the VLA searches commensally for narrowband signals

drifting in frequency, beamforming toward Gaia-catalogue stars in the field of view (Myburgh et al. 2026) — a design that assumes a persistent, Doppler-drifting carrier, precisely the signature a rarely and directionally communicating lineage would not produce.

Fourth, interstellar objects deserve scrutiny as possible **messenger probes** (Bracewell 1960). The prior probability may be low, but the scientific value of discriminating natural from artificial interstellar objects is high.

Fifth, technosignature theory should model **governed behavior**. Not all capable civilizations maximize visibility; some may minimize interference, leakage, or detectable alteration, and a search theory that assumes maximal conspicuousness will systematically miss them.

Sixth, the Solar System itself should not be excluded. Quiet probes, if any exist nearby, would likely be small, cold, and dormant. Searches should be careful, not sensational: artificiality requires evidence, and natural explanations should dominate until displaced. But the hypothesis is not logically absurd. Watters et al. (2026) supply a worked example of this discipline: re-examining claims of anomalous, possibly artificial features in first-epoch Palomar photographic plates, they show the reported deficit near Earth’s shadow and the correlation with nuclear-test dates both dissolve once the background feature distribution is modeled correctly — a caution that applies as much to future searches for the quiet-probe signatures this paper proposes as to the claims it corrects.

Finally, the absence of obvious probes should be interpreted against explicit models. It is not enough to say “we do not see them.” We need to specify what signatures a given architecture predicts, at what intensity, in what locations, and under what governance rules. Fermi reasoning without detectability modeling is incomplete.

11. Discussion

The architecture developed in this series complicates both optimistic and pessimistic interpretations of the Fermi Paradox.

Against naive optimism, it shows that self-replication is not a magic wand. The hard problem is not launching one probe. It is maintaining a supercritical autonomous lineage under failure, closure, braking, resource, governance, and mission-integrity constraints. The “one probe fills the Galaxy” argument holds only after these conditions are satisfied — and the series’ computation suggests they are satisfied only narrowly, if at all.

Against naive pessimism, it shows that absence of obvious megastructures or broadcasts does not decisively rule out self-replicating systems. A probe lineage optimized for knowledge preservation may be quiet by design. It may avoid inhabited systems. It may suppress broad transmissions. It may use small-body reservoirs and narrow-beam communication. It may be present in forms that current searches are not optimized to detect.

The key tension is this: quietness can explain non-detection only up to a point. A single quiet node is easy to miss. A million quiet nodes are harder to miss. A billion quiet nodes, distributed over Galactic time, should produce some anomalies unless detectability is extremely low or governance strongly suppresses expansion into observable regimes. Thus the quiet-archive hypothesis is not a free escape, in the way many catalogued Fermi solutions implicitly are (Webb 2015). It must make predictions.

The discussion above treats the quiet-archive hypothesis mainly as an engineering and detection problem, but the silence itself supports a further, more uncomfortable reading. Section 9's value-stability filter allows that launched lineages may "mutate into forms that self-limit, fragment, or fail"; Section 2's severity argument holds that even one reliable lineage, given Galactic time, should already be visible. Together these suggest that the absence of any visible lineage may be inductive evidence that this exact undertaking — launching a self-replicating, value-bearing probe into deep time — tends to end badly wherever it has been tried, and that we cannot observe or learn from those prior failures before repeating the experiment ourselves. This is a distinct concern from the existential-risk and backup-of-humanity arguments raised in the ethics paper: it is not a claim that launching endangers us, but that launching may be joining a pattern of failure we have no way to inspect. The same silence keeps open a second possibility, which Sections 2 and 9 both name via Sagan and Newman's (1983) objection: that non-launch itself is the outcome the silence is evidence for, and that the rational and ethical response to this architecture's own risk profile is not a more restrained probe but no probe at all. That option sits outside the dual-use framing this series otherwise favors, in which R_{eff} -bounding governance is engineered into a lineage that gets built regardless; it is a prior question about whether building should proceed.

Governance is also not ethically free-standing; Section 4 shows viability, not just per-stage success probability, setting R_{eff} , and Section 6 notes that classifying a system as contact-restricted or settlement-prohibited lowers that viability directly. A more protective governance posture therefore pushes computed R_{eff} toward the subcritical-silence regime named in Section 7, where the lineage dies out regardless of how sound its ethics are — turning "how restrictive should the immutable core be" from an abstract values question into a numeric trade against the lineage's own persistence. The trade cuts both ways. Section 7's ethical self-limitation regime, in which governance deliberately holds R_{eff} near or below one, sits in direct tension with Section 6's own warning that an overly restrictive lineage "may fail to aid civilizations it could have helped" and "may decay into a passive archive rather than an active lineage." The architecture names both horns of this dilemma without resolving them: restraint too weak risks the predatory or hostile expansion regime of Section 7; restraint too strong risks trading the lineage's own future, and its usefulness to others, for a safety margin it may not need.

Two further asymmetries follow from the same governance logic. First, Section 7's predatory or hostile expansion regime — R_{eff} above one, high or variable D , weak noninterference — implies that some lineage, somewhere, could be prey rather than predator. A lineage built by Section 6's rules to minimize its own detectability and contact footprint has not thereby been built to recognize or respond to a different

lineage’s predation; restraint and self-defense are not the same design target, and nothing in this framework asks whether the quiet-archive architecture retains any capacity for the latter. Second, the governance posture Section 6 treats as individually rational and ethical for any one lineage — low D , no broad broadcasting, contact only on reciprocal terms — is exactly the posture that, adopted independently by many civilizations for the same good reasons, could produce a self-reinforcing galactic norm of mutual isolation. Each participant’s restraint is locally justified by its own risk calculus; the aggregate outcome is a Galaxy in which civilizations that could have found or helped one another do not, including some that go undetected, unaided, and possibly extinct. This is a collective-action problem distinct from the single-lineage contact-ethics question the governance paper addresses: it does not ask whether one probe should make contact, but what follows when the same correct answer to that question is reached everywhere at once.

The most useful outcome of this paper is therefore methodological. It moves the debate from binary speculation to parameterized search theory. For any proposed probe architecture, ask: What is R_{eff} ? What is D ? What is G ? What signatures follow? And what observations would constrain them? Only after those questions are answered can the Fermi Paradox be applied sharply, rather than invoked as a slogan.

12. Conclusion

A slow, self-repairing, self-replicating interstellar AI probe designed to preserve knowledge across deep time changes the Fermi Paradox without eliminating it. It supports the classical insight that replication, not speed, is the decisive driver of galactic reach. If such lineages are reliable and supercritical, they can spread across the Galaxy on timescales short compared with its age. But it also changes what we should expect to see. A deep-time archive lineage may be quiet, cold, sparse, ethically constrained, and hidden in small-body reservoirs. It may communicate by narrow beams and messenger probes rather than broad broadcasts. It may avoid inhabited worlds. It may treat knowledge preservation, not contact or expansion, as its terminal purpose.

The resulting framework has three parameters: the effective reproduction number R_{eff} , the detectability D , and the governance posture G . Classical Fermi reasoning occupies the high- R_{eff} , high- D , expansionist- G regime. The architecture developed here occupies a more elusive regime: potentially supercritical reproduction, low detectability, and restrictive noninterference. If we do not see such systems, the filters may lie in reproduction reliability, industrial closure, braking, value stability, governance, or our failure to search for the right signatures — and the series’ own demographics suggest the reproduction filter alone may be decisive.

The question “Where is everybody?” remains. But for the class of systems considered here, a sharper question is required: what would a quiet, million-year-old machine built to preserve knowledge — not to announce itself — actually look like? Until we can answer that, the absence of visible galactic civilizations is not yet the absence of deep-time intelligence.

References

- Andrews, D. G., & Zubrin, R. M. (1990). Magnetic sails and interstellar travel. *Journal of the British Interplanetary Society*, 43, 265–272.
- Bostrom, N. (2014). *Superintelligence: Paths, Dangers, Strategies*. Oxford University Press.
- Bracewell, R. N. (1960). Communications from superior galactic communities. *Nature*, 186, 670–671.
- Burleigh, S., Hooke, A., Torgerson, L., Fall, K., Cerf, V., Durst, B., Scott, K., & Weiss, H. (2003). Delay-tolerant networking: An approach to interplanetary Internet. *IEEE Communications Magazine*, 41(6), 128–136.
- Cerf, V., Burleigh, S., Hooke, A., Torgerson, L., Durst, R., Scott, K., Fall, K., & Weiss, H. (2007). *Delay-Tolerant Networking Architecture* (RFC 4838). Internet Engineering Task Force.
- Ćirković, M. M. (2009). Fermi’s paradox — The last challenge for Copernicanism? *Serbian Astronomical Journal*, 178, 1–20.
- Crawford, I. A. (2000). Where are they? Maybe we are alone in the Galaxy after all. *Scientific American*, 283(1), 38–43.
- Curtis, O., Rowland, A. J., Wright, J. T., Gronwall, C., Helton, J. M., & Leja, J. (2026). The $\tilde{G}$ infrared search for extraterrestrial civilizations with large energy supplies. V. When galaxies glow with industry. *arXiv preprint* arXiv:2608.12458.
- Freitas, R. A. (1980). A self-reproducing interstellar probe. *Journal of the British Interplanetary Society*, 33, 251–264.
- Freitas, R. A., & Merkle, R. C. (2004). *Kinematic Self-Replicating Machines*. Landes Bioscience.
- Haqq-Misra, J., & Baum, S. D. (2009). The sustainability solution to the Fermi paradox. *Journal of the British Interplanetary Society*, 62, 47–51.
- Harris, T. E. (1963). *The Theory of Branching Processes*. Springer.
- Hart, M. H. (1975). An explanation for the absence of extraterrestrials on Earth. *Quarterly Journal of the Royal Astronomical Society*, 16, 128–135.
- Ivliev, S. (2026). Autonomous AI-cosmoindustry and the quiet expansion filter: A threshold-based resolution of the Fermi paradox. *arXiv preprint* arXiv:2606.13914.
- Lam, C., Ballard, S., Sagar, S., & Daniel, K. J. (2026). An increase in the Galactic planet-host fraction fails to reproduce the Galactic height trend in planet occurrence. *The Astronomical Journal*, 172, 143.
- Likavčan, L. (2026). The grass of the universe: Rethinking technosphere, planetary history, and sustainability with Fermi paradox. *Acta Astronautica*. <https://doi.org/10.1016/j.actaastro.2026.08.004>

- Merkle, R. C. (1987). A digital signature based on a conventional encryption function. In *Advances in Cryptology — CRYPTO '87* (pp. 369–378). Springer.
- Myburgh, T., Stiegler, N., Tremblay, C. D., Bright, J. S., & Donnachie, R. A. (2026). The VLA and high-frequency SETI: Expanding the search for life. *Acta Astronautica*. <https://doi.org/10.1016/j.actaastro.2026.08.032>
- Omohundro, S. M. (2008). The basic AI drives. In *Proceedings of the First Conference on Artificial General Intelligence (AGI 2008)* (pp. 483–492). IOS Press.
- Sagan, C., & Newman, W. I. (1983). The solipsist approach to extraterrestrial intelligence. *Quarterly Journal of the Royal Astronomical Society*, 24, 113–121.
- Tipler, F. J. (1980). Extraterrestrial intelligent beings do not exist. *Quarterly Journal of the Royal Astronomical Society*, 21, 267–281.
- von Neumann, J., & Burks, A. W. (1966). *Theory of Self-Reproducing Automata*. University of Illinois Press.
- Watters, W. A., Dominé, L., Little, S., Pratt, C., Knuth, K. H., & Szenher, M. (2026). Critical evaluation of studies alleging evidence for technosignatures in the POSS1-E photographic plates. *Publications of the Astronomical Society of Australia*. <https://doi.org/10.1017/pasa.2026.10230>
- Webb, S. (2015). *If the Universe Is Teeming with Aliens... Where Is Everybody? Seventy-Five Solutions to the Fermi Paradox and the Problem of Extraterrestrial Life* (2nd ed.). Springer.
- Wright, J. T. (2018). Exoplanets and SETI. In H. J. Deeg & J. A. Belmonte (Eds.), *Handbook of Exoplanets*. Springer.
- Zuckerman, B., & Hart, M. H. (Eds.). (1995). *Extraterrestrials: Where Are They?* (2nd ed.). Cambridge University Press.